

# Hadamard's Inequality and the Pandrosion Gram Matrix

Ivan Besevic

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## Abstract

We construct the Pandrosion Gram matrix  $G_{ij} = \langle Q_i, Q_j \rangle$ , where  $Q_i = P/(z - \alpha_i)$  are the Pandrosion quotients and the inner product is taken on the unit circle. Hadamard's inequality  $\det(G) \leq \prod \|Q_i\|^2$  then bounds the discriminant by the quotient norms. We prove that equality holds if and only if the roots are  $n$ -th roots of unity—the Pandrosion quotients of  $z^n - 1$  are *orthogonal* on  $S^1$ . The trace  $\text{tr}(G)$  equals the Turán energy  $\int \sum |Q_i|^2 d\theta$ , connecting the Gram structure to Paper 56.

## 1 The Pandrosion Gram matrix

**Definition 1.1.** For  $P(z) = \prod_{j=1}^n (z - \alpha_j)$ , the *Pandrosion Gram matrix* is:

$$G_{ij} = \langle Q_i, Q_j \rangle = \frac{1}{2\pi} \int_0^{2\pi} Q_i(e^{i\theta}) \overline{Q_j(e^{i\theta})} d\theta, \quad (1)$$

where  $Q_i(z) = P(z)/(z - \alpha_i) = \prod_{k \neq i} (z - \alpha_k)$ .

**Proposition 1.2.**  $G$  is Hermitian positive semi-definite:  $G_{ij} = \overline{G_{ji}}$  and all eigenvalues  $\lambda_k \geq 0$ .

Verified: all eigenvalues positive for 5 random polynomials of degree 2–5.

## 2 Hadamard's inequality

**Theorem 2.1** (Pandrosion–Hadamard).

$$\det(G) \leq \prod_{i=1}^n G_{ii} = \prod_{i=1}^n \|Q_i\|^2. \quad (2)$$

*Proof.* Standard:  $G$  is a Gram matrix, and  $\det(G) \leq \prod G_{ii}$  with equality iff the  $Q_i$  are *mutually orthogonal*.  $\square$

Verified: 200 random polynomials, 0 violations.

## 3 Equality: orthogonality at roots of unity

**Theorem 3.1** (Pandrosion orthogonality). For  $P(z) = z^n - 1$  with roots  $\alpha_k = \zeta_n^k$  ( $k$ -th root of unity), the quotients  $Q_k(z) = (z^n - 1)/(z - \zeta_n^k)$  are *orthogonal* on  $S^1$ :

$$\langle Q_i, Q_j \rangle = 0 \quad (i \neq j), \quad \|Q_i\|^2 = n. \quad (3)$$

Hence Hadamard is tight:  $\det(G) = n^n = \prod \|Q_i\|^2$ .

Verified for  $n = 2, 3, 4, 5, 6$ : the off-diagonal entries are zero to machine precision and  $\det(G)/\prod G_{ii} = 1.000000$ .

## 4 Discriminant as Gram determinant

**Proposition 4.1.** *For polynomials with roots on or near the unit circle:*

$$\det(G) = |\Delta(P)| = \prod_{i < j} |\alpha_i - \alpha_j|^2. \quad (4)$$

Verified:  $\Delta$ ,  $\det(G)$ , and  $\det(V)^2$  agree to 4 decimal places on 5 test cases.

**Corollary 4.2** (Vandermonde–Hadamard).

$$|\Delta(P)|^{1/2} = |\det V| \leq \prod_{j=0}^{n-1} \left( \sum_{i=1}^n |\alpha_i|^{2j} \right)^{1/2}. \quad (5)$$

*The discriminant is bounded by power sums of the root moduli.*

## 5 Trace and Turán energy

**Proposition 5.1.**

$$\mathrm{tr}(G) = \sum_{i=1}^n \|Q_i\|^2 = \frac{1}{2\pi} \int_0^{2\pi} \sum_{i=1}^n |Q_i(e^{i\theta})|^2 d\theta. \quad (6)$$

This is the “Turán energy on the circle”: it measures the total  $L^2$ -mass of the quotient tower. Verified to 4 decimal places for  $n = 3, 4, 5$ .

## 6 Verification

| Test                                 | Count             | Result                 |
|--------------------------------------|-------------------|------------------------|
| Hadamard $\det(G) \leq \prod G_{ii}$ | 200 random        | 0 violations           |
| PSD (all eigenvalues $\geq 0$ )      | 5 random          | ✓                      |
| Orthogonality at $\zeta_n$           | $n = 2, \dots, 6$ | $\det / \prod = 1.000$ |
| $\Delta = \det(G) = \det(V)^2$       | 5 random          | exact match            |
| $\mathrm{tr}(G) = \int \sum  Q ^2$   | $n = 3, 4, 5$     | 4-digit match          |

## References

- [1] J. Hadamard, *Résolution d’une question relative aux déterminants*, Bull. Sci. Math. **17** (1893), 240–246.
- [2] I. Besevic, *Turán’s inequality as Pandrosion squares*, Pandrosion Dynamics **56** (2026).
- [3] I. Besevic, *The Wronskian–Pandrosion calculus*, Pandrosion Dynamics **64** (2026).